

Informational Power and Institutional Design: When a Hegemon May Choose Consensus

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2026-04-27

Abstract

When does a powerful state prefer consensus to majority rule in an international organization? I develop a formal model of institutional design in which a hegemon has private information about the value of multilateral cooperation. Under majority rule, weaker states can form winning coalitions without the hegemon, so its private information has limited strategic value. Under unanimity, every agreement requires the hegemon's approval, forcing weaker states to decide how much to offer without knowing how valuable cooperation really is. This uncertainty works in the hegemon's favor: weaker states pay more than necessary to secure agreement, and majority rule eliminates this advantage entirely. This screening advantage means that, once cooperation is promising enough for weaker states to participate under unanimity, the hegemon strictly prefers it. Below that threshold, majority can dominate—but only because it makes participation easier, not because it gives better terms. Consensus, in this account, is not a concession to weaker states but an institutional arrangement that makes informational advantage politically productive—offering a distributive explanation for consensus rules in organizations such as the WTO.

1 Introduction

Most international organizations operate by consensus, even when powerful states drive their creation (Gould 2022). The international arena is where material asymmetries should matter most relative to domestic politics. If institutional rules reflect power, powerful states might be expected to favor arrangements that let them turn that power into a distributive advantage. Why would a powerful state ever prefer consensus over majority rule, where it could use agenda power to impose its preference without any state having veto power?

In bargaining, majority rule allows a powerful actor to form coalitions, exclude rivals, and keep the surplus (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Consensus does the opposite. By giving every participant a veto, it forces powerful states to surrender the very tools they might value most (Keohane 1984). Yet international organizations, such as the GATT/WTO, are built around consensus (Steinberg 2002; Koremenos, Lipson, and Snidal 2001).

Existing answers do not fully resolve the puzzle. A common response claims that powerful states still influence outcomes through informal agenda power, even under consensus (Steinberg 2002; Stone 2011). Yet this answer is conceptually unsatisfying: under unanimity, all participants are veto players, so informal agenda influence cannot enable exclusion and surplus capture as majority rule does. More broadly, existing accounts either see consensus as constraining powerful states—a mechanism of institutional self-binding that limits the hegemon’s own discretion (Keohane 1984; Ikenberry 2001)—or as a concealment for conventional power, where formal equality masks asymmetric influence (Steinberg 2002; Gruber 2000). Neither explains when and why a hegemon would prefer unanimity to a rule that allows exclusion.

This paper theorizes when and why a hegemon may prefer unanimity to majority rule in international organizations. The central claim is that consensus is an institutional technology of power. I show that consensus can make informational power more politically productive than the coalition arithmetic of majority rule. The key institutional choice is between two distinct modes of exercising power: agenda power and informational power. By informational power I mean the bargaining advantage that a privately informed actor derives from being pivotal: when other players must secure its approval without knowing its type, uncertainty itself becomes a source of rent.

To formalize the argument, I develop a model in which a hegemon chooses between majority and unanimity. Proposal rights are symmetric under both rules, isolating the effect of the voting rule itself. The hegemon privately observes the value of cooperation before weaker states decide whether to enter. The central difference is that the two rules reward this informational advantage differently. Under majority rule, weak states can form coalitions without making the hegemon pivotal, eliminating the screening problem. Under unanimity, weak states must include the hegemon in every proposal and decide how much to offer without knowing the probability of acceptance. This screening generates a cutoff at which weak proposers switch from aggressive to conservative treatment of the hegemon, giving the hegemon a strictly higher conditional payoff than under majority at every level of uncertainty. The institutional comparison reduces to a single distinction: unanimity dominates wherever it can sustain entry, and majority’s only advantage is wider institutional viability.

The substantive implication is that consensus is most valuable to a hegemon when the prospects of multilateral cooperation are promising enough that weaker states are willing to come to the table. Under those conditions, unanimity amplifies the hegemon’s informational advantage. When cooperation looks too uncertain for weaker states to participate at all, majority rule can be preferred—not because it gives the hegemon better bargaining terms, but because it lowers the bar for participation.

2 A Motivating Example

Before turning to the general model, consider a simplified special case with three players and one bargaining round to build intuition for the mechanism. The general model (Section 3) extends this to N players, two rounds of Baron-Ferejohn bargaining, and entry costs. There is one hegemon H and two weak states W_1, W_2 . The value of cooperation is either $V(0) = 1$ (low) or $V(1) = 2$ (high); H observes the state, the weak states do not. Each weak state’s outside option is 0; H ’s outside option is $\alpha V(\theta)$ with $\alpha = 0.2$. A proposer is selected uniformly at random; to illustrate the screening mechanism, I focus on the case where a weak state is selected and makes a take-it-or-leave-it offer specifying H ’s share y_H . If H rejects, each player receives their disagreement payoff.

Throughout the paper, I model consensus as unanimity over a binary accept/reject decision. This

abstracts the distinction between unanimity (everyone must explicitly consent to an agreement) and consensus (no one explicitly blocks a deal), but captures the central strategic feature: any member can block agreement.

Under majority rule, W_1 and W_2 can pass a proposal without H 's vote. The proposing weak state excludes H from the coalition, and H captures $0.2V(\theta)$ from its bilateral alternative regardless of what W believes. The hegemon's expected payoff is $0.2 + 0.2\mu$, where $\mu = \Pr(\theta = 1)$. The payoff is linear in beliefs, with no strategic discontinuity. Because the hegemon's vote is never needed, its private information creates no leverage.

Under unanimity, H 's vote is required, and the proposing weak state faces a dilemma. It must decide how much to offer H without knowing θ :

- An aggressive offer ($y_H = 0.2$) pays the hegemon its low-state outside option. Type $\theta = 0$ is indifferent and accepts; type $\theta = 1$ rejects, since it wants 0.4. The weak proposer gets 0.8 if $\theta = 0$ and 0 if $\theta = 1$.
- A conservative offer ($y_H = 0.4$) pays the hegemon its high-state outside option. Both types accept. The weak proposer gets 0.6 if $\theta = 0$ and 1.6 if $\theta = 1$.

The weak state prefers the aggressive offer when $0.8(1 - \mu) > 0.6(1 - \mu) + 1.6\mu$, which simplifies to $\mu < 1/9$. This defines a screening cutoff: the belief at which the weak proposer switches from aggressive to conservative treatment of the hegemon.

The switch in behavior creates a discrete jump in the hegemon's expected payoff. Below the cutoff, the hegemon earns $0.2 + 0.2\mu$, the same as under majority. Above it, the expected payoff jumps to 0.4, constant and independent of μ . The source of the jump is overpayment: on the conservative branch, type $\theta = 0$ receives 0.4 but would accept 0.2. Weak states cannot avoid this cost because they must secure the hegemon's vote without knowing the type. At the cutoff $\mu^* = 1/9$, the jump is $8/45 \approx 0.18$, about 16% of expected surplus.

The screening jump gives the hegemon a strictly higher conditional payoff under unanimity than under majority for every belief above the cutoff. The general model (Section 3 onwards) shows this holds across the entire belief space (Theorem~1).

The general model enriches this logic in several respects. With N players and two-round Baron-

Ferejohn bargaining with discounting ($\beta < 1$), the richer environment produces two screening cutoffs (one per round) and an entry threshold that creates a second channel through which the voting rule matters. Theorem~1 shows that the pattern identified here generalizes: conditional on entry, unanimity gives the hegemon a strictly higher payoff at every level of uncertainty. The institutional classification (Proposition~4) establishes that unanimity dominates wherever it can sustain entry; majority dominates only through wider institutional viability.

3 The Model

Definition 1 (Institutional Design Game). *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $N - 1$ weak states $W_1, \dots, W_{N-1}$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ ; the weak states do not.*
- *Disagreement payoffs: each W receives 0;¹ H receives $\alpha V(\theta)$, with $\alpha \in (0, 1/r)$.²*
- *The common discount factor is $\beta \in (0, 1)$.*
- *Each weak state that joins the institution pays an entry cost $c > 0$.*
- *The voting rule is $R \in \{M, U\}$: under majority (M), decisions pass with $q = \lfloor N/2 \rfloor + 1$ votes; under unanimity (U), decisions require all N votes.³ Both rules have symmetric proposal rights (each player recognized with probability $1/N$).*

¹The normalization $d_W = 0$ is without loss of generality: it requires only that weak states' outside options are symmetric and strictly lower than the hegemon's. The pie $V(\theta)$ is then interpreted as the surplus above weak states' common outside option.

²The proportional form $d_H = \alpha V(\theta)$ captures bilateral alternatives that scale with the value of multilateral cooperation—the same factors that make multilateral cooperation valuable also improve the hegemon's outside option. The restriction $\alpha < 1/r$ ensures bilateral alternatives are strictly dominated by multilateral cooperation for all states of the world.

³In this game, each player's action is binary: accept or reject. There is no abstention or silence option. The institutional distinction between consensus—where silence is treated as tacit approval—and formal unanimity—where explicit approval is required—therefore does not arise: with binary actions $\{\text{accept, reject}\}$, the two rules are equivalent. I use the terms interchangeably. In practice, “consensus” at the WTO refers to the norm of not calling a formal vote, which approximates unanimity when any member can block by objecting.

Let $V_e(\mu) \equiv 1 + \mu(r - 1)$ denote the expected value of cooperation at posterior belief μ , and define $x \equiv (N - 1)\alpha r$ as a notational shorthand.

The two-round bargaining structure follows the Baron-Ferejohn framework that is standard in legislative bargaining. A single round would reduce the game to a one-shot ultimatum, and one might question whether the screening mechanism depends on that special structure. Two rounds ensure that rejection in Round 1 leads to a continuation game—not immediate disagreement—so the hegemon’s reservation value in Round 1 reflects a credible outside option within the institution. The second round also generates off-path belief updating: when the high type rejects an aggressive offer in Round 1, weak states learn $\theta = 1$ with certainty, which pins down the Round 2 offers and, by backward induction, the Round 1 screening cutoff. Beyond robustness, the two-round structure generates analytically richer results: the R1 screening cutoff is independent of α (Proposition 2), and the two screening cutoffs (one per round) create a richer conditional payoff advantage for the hegemon. The motivating example (Section 2) shows that the mechanism may operate in one round; the two-round structure confirms that it survives in the richer Baron-Ferejohn environment.

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Entry).** Nature draws θ ; the hegemon observes θ . Weak states hold prior belief $p = \Pr(\theta = 1)$ and simultaneously decide whether to enter (paying cost c). The hegemon participates automatically—it has chosen the institution at Stage 0 and faces no entry cost. The institution forms if and only if all $N - 1$ weak states enter; partial entry is not permitted.⁴ Since weak states are ex ante identical, I restrict attention to the symmetric entry equilibrium: all $N - 1$ weak states enter if and only if the expected equilibrium payoff satisfies $V_W^{R1}(p, R) \geq c$; otherwise no institution forms and each player receives their disagreement payoff.
3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn

⁴The all-or-nothing entry technology is a simplification that avoids a separate extensive form for coalition formation at the entry stage. It is appropriate when the institution requires broad membership to function (as in a multilateral trade round) or when the voting rule is defined over a fixed membership. The parameter N should be read as the size of the institution the hegemon proposes to create, not the total number of states in the international system. The hegemon invites a specific group of $N - 1$ weak states; the entry decision is whether to join this particular institution.

bargaining game over a pie of size $V(\theta)$:

- **Round 1:** a proposer is selected uniformly at random (probability $1/N$). The proposer offers a division. All members vote according to rule R . If accepted, payoffs are realized. If rejected, the game proceeds to Round 2 with payoffs discounted by β .
- **Round 2 (terminal):** a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, payoffs are realized. If rejected, each player receives their disagreement payoff.

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs satisfy sequential rationality.⁵

Throughout the game, the hegemon's information sets are singletons: H knows θ at every decision node. Each weak state's information sets pool the two branches of θ : at any decision node, W_i knows the history of proposals and responses, but not θ . Weak states hold belief μ over θ , which equals the prior p at the entry stage and may be updated during bargaining (e.g., after a rejection in Round 1). This asymmetry is what generates the screening problem when a weak state proposes under unanimity.

Figures 1 and 2 present the extensive form of the game: Figure~1 covers institutional choice and entry (Stages 0–1); Figure~2 details the bargaining subgame under unanimity (Stage 2).

⁵Tie-breaking convention: when H is indifferent between accepting and rejecting an offer (offer equals reservation value), H accepts.

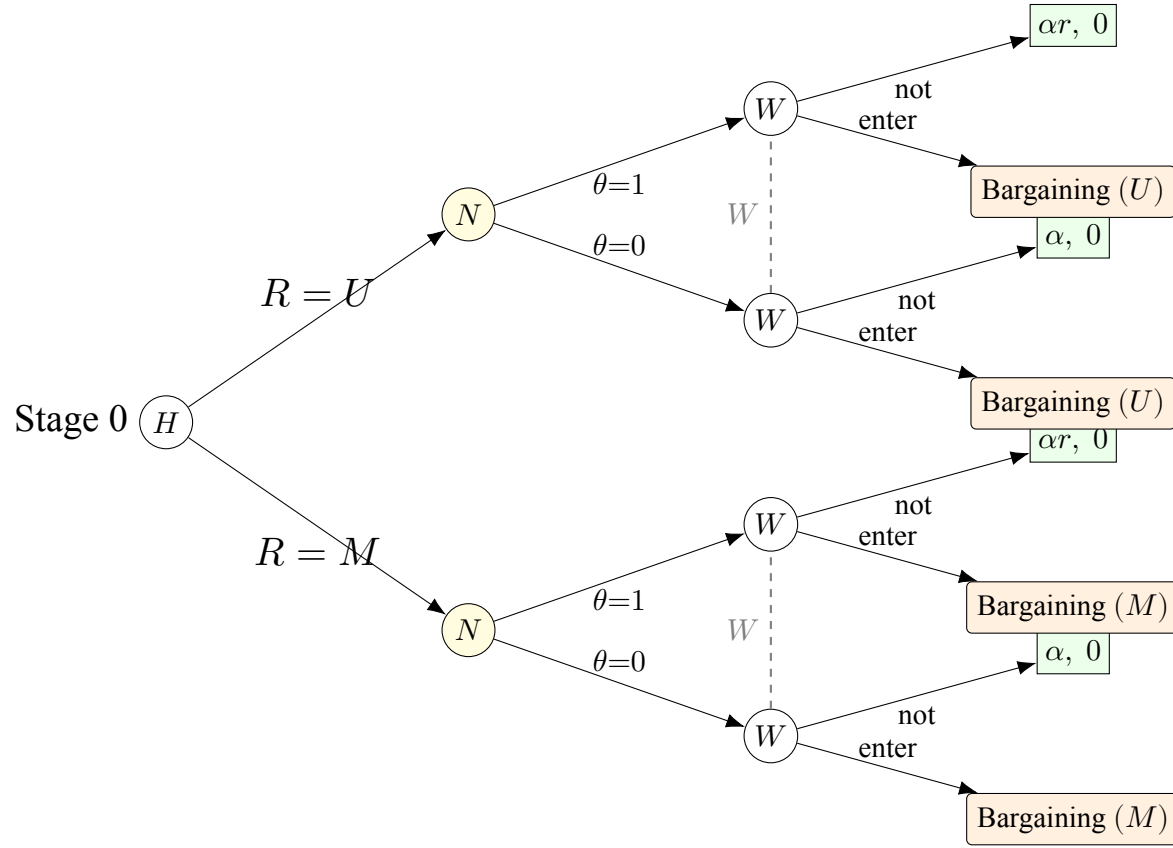


Figure 1: Extensive form: Stages 0–1 (institutional choice and entry). At Stage 0, H selects voting rule $R \in \{M, U\}$. Nature draws $\theta \in \{0, 1\}$; H observes θ , weak states hold prior $p = \Pr(\theta=1)$ and decide entry (cost c). Dashed lines: W 's information sets (W does not observe θ). Non-entry payoffs: $(u_H, u_W) = (\alpha V(\theta), 0)$.

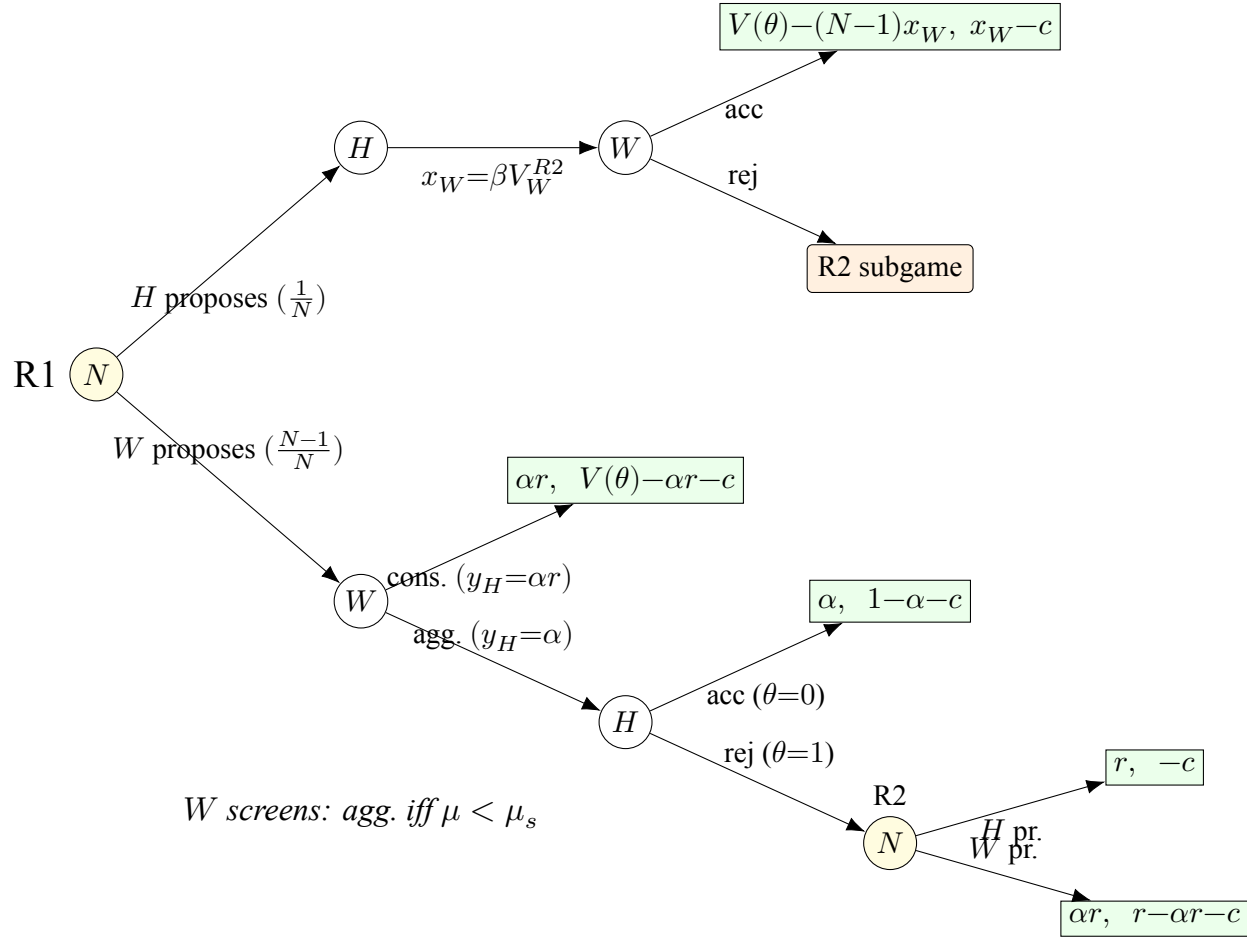


Figure 2: Extensive form: Stage 2 (bargaining under unanimity, Rounds 1-2). Payoffs are (u_H, u_W) . A proposer is selected uniformly ($1/N$). When W proposes, W chooses between an aggressive offer that only $\theta=0$ accepts and a conservative offer that both types accept. Under the aggressive offer, $\theta=1$ rejects, revealing the state and leading to R2 (payoffs discounted by β). In R2, W knows $\theta=1$ (complete information); W offers αr , H offers 0; disagreement yields $(\alpha r, -c)$. When H proposes in R1, H offers βV_W^{R2} to each W ; W accepts in equilibrium. Offer labels and terminal payoffs show R2 values (where H 's reservation is $\alpha V(\theta)$) for readability; exact R1 offer levels are $\beta V_H^{R2}(\theta=0, \mu'=1)$ (aggressive) and $\beta V_H^{R2}(\theta=1)$ (conservative), derived in Appendix A.3. Under majority rule, H 's vote is not pivotal, eliminating the screening mechanism (Proposition 1).

4 Majority Rule: Linear Bargaining Without Screening

Under majority rule, a proposal passes with $q = \lfloor N/2 \rfloor + 1$ votes. The key structural feature is that weak states can form a winning coalition without including the hegemon. When a weak state proposes, it assembles a majority from the other weak states alone; the hegemon, excluded from the coalition, captures its bilateral alternative $\alpha V(\theta)$ regardless of what weak states believe about the state of the world.

Proposition 1 (Majority rule produces no screening). *Under majority rule with symmetric proposal rights, the hegemon's expected continuation value $E_\theta[V_H^{R1}(\theta, \mu, M)]$ is affine in posterior beliefs μ . There is no screening cutoff.*

Proof. See Appendix B.1. □

Because the hegemon's vote is never needed, weak states never face a choice between offers that depend on inferring the hegemon's type. Majority transforms bargaining into coalition arithmetic: the proposer buys the cheapest votes, and the hegemon is simply paid its outside option. The hegemon's private information affects the value of the agreement but creates no strategic discontinuity and no screening rent.

5 Consensus: Screening Through Pivotal Inclusion

Under unanimity, every proposal requires the approval of all members, including the hegemon. When a weak state proposes, it must secure H 's vote without knowing whether the pie is high or low. As in the motivating example (Section 2), this creates a screening problem: the proposer must choose between an aggressive offer that only the low type accepts and a conservative offer that both types accept. The conservative offer overpays the low type—the price of uncertainty.

This screening logic operates in both bargaining rounds. In each round, the weak proposer's indifference between aggressive and conservative offers defines a screening cutoff: a belief threshold above which the proposer switches from aggressive to conservative treatment. The R2 cutoff has a closed form in all cases; the R1 cutoff has a closed form when the hegemon's outside option is moderate (Appendix A).

Proposition 2 (R1 screening cutoff under consensus). *Under unanimity, there exists a unique screening cutoff $\mu_s^{R1} \in (0, 1)$ such that the weak proposer prefers the aggressive offer for $\mu < \mu_s^{R1}$ and the conservative offer for $\mu > \mu_s^{R1}$. When $\alpha < \bar{\alpha}(r, \beta, N)$, the cutoff takes the closed form:*

$$\mu_s^{R1} = \frac{\phi - \sqrt{\phi^2 - 4(N-2)}}{2(N-2)}, \quad \phi \equiv \frac{rN - \beta(N-1+r)}{\beta(r-1)}, \quad (1)$$

and is independent of α . The regime threshold is

$$\bar{\alpha}(r, \beta, N) \equiv \frac{\mu_s^{R1} \cdot r}{r-1 + \mu_s^{R1}}, \quad (2)$$

where μ_s^{R1} is evaluated from (1). For $\beta = 1$, the cutoff simplifies to $\mu_s^{R1} = 1/(N-2)$.⁶

Proof. See Appendix B.2. □

The independence of the screening cutoff from the hegemon's outside option strength—which holds when $\alpha < \bar{\alpha}$ —is substantively important. What determines how aggressively weak states treat the hegemon is not how valuable bilateral alternatives are, but how consequential the hegemon's private information is for the value of cooperation. The cutoff also depends on bargaining patience and the number of states that must agree.

Proposition 3 (Screening creates an informational rent). *Under unanimity, the hegemon's expected continuation payoff in R1 has an upward jump at μ_s^{R1} :*

$$\text{Jump} = (1 - \mu_s^{R1}) \cdot \frac{(N-1)\beta(r-1)}{N^2} > 0. \quad (3)$$

Proof. See Appendix B.3. □

At the screening cutoff, weak proposers switch from treating the hegemon as the low type to hedging against the possibility of the high type. The conservative offer pays the low type as if it could be the high type, creating a discrete upward shift in the hegemon's expected payoff. The screening rent is

⁶When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch and depends on α (Appendix A.5). The cutoff remains unique and the screening jump remains positive. The proof of Theorem 1 (Appendix B.5) covers both regimes explicitly: the payoff decomposition is additive in the R1 and R2 corrections regardless of the relative ordering of μ_s^{R1} and μ_s^{R2} , and the endpoint argument establishes $D(\mu) > 0$ on all branches. The Corollary and Proposition then follow from Theorem 1 without further dependence on the regime.

largest when the hegemon's private information is most consequential for the value of cooperation (large r), when bargaining is patient enough to make the continuation threat credible (large β), and in smaller organizations, where the per-member screening advantage $(N - 1)/N^2$ is larger (the expression is monotonically decreasing in N for $N \geq 2$).

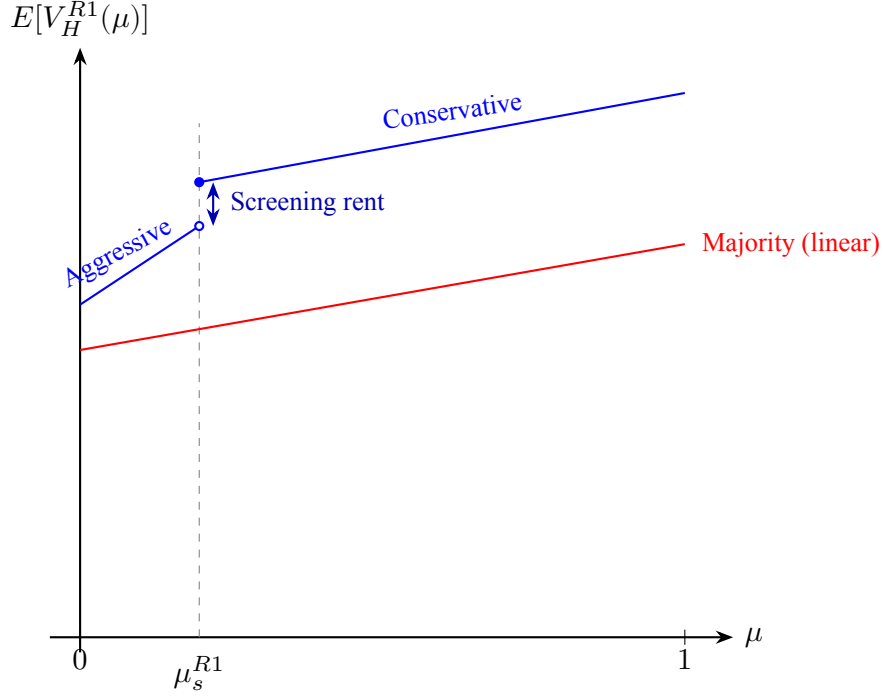


Figure 3: Schematic of the hegemon's expected R1 payoff as a function of posterior belief μ , based on Example 1 parameters ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$). Under majority (red), the payoff is linear in μ with no discontinuity. Under unanimity (blue), the payoff has two branches separated by an upward jump at the screening cutoff $\mu_s^{R1} \approx 0.197$: to the left, weak proposers play aggressively; to the right, they switch to conservative offers that overpay the low type. The gap is the screening rent that drives the hegemon's preference for unanimity (Theorem 1).

Example 1 (Screening jump: concrete magnitudes). Let $N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$.⁷ The R1 screening cutoff is $\mu_s^{R1} \approx 0.197$. At beliefs just below the cutoff, weak proposers play

⁷The parameters are illustrative but grounded in plausible magnitudes. $N = 5$ represents a small plurilateral negotiation (e.g., the Quad in early GATT rounds). $r = 1.5$ implies that the high-value state of cooperation is 50% more valuable than the low-value state, consistent with the range of estimated welfare gains across ambitious versus modest trade liberalization scenarios. $\alpha = 0.3$ means the hegemon's bilateral alternative captures about a third of multilateral value—substantial but not dominant. $\beta = 0.9$ reflects moderate discounting—rejection is costly but not catastrophic—standard in Baron-Ferejohn applications.

aggressively and the hegemon's expected payoff is $E[V_H^{R1}] \approx 0.544$. Just above the cutoff, the payoff jumps to ≈ 0.602 —an increase of about 5.3% of expected surplus. Under majority at the same belief, $E[V_H^{R1}] \approx 0.428$, with no jump anywhere. The screening mechanism gives the hegemon 27% more than majority on the aggressive branch and 41% more on the conservative branch. The discrete gap between branches is the screening rent that drives the hegemon's preference for unanimity.

6 Entry and Institutional Viability

Having established that unanimity generates a conditional payoff advantage for the hegemon (Propositions 2–3), I now turn to the entry margin—the channel through which majority can nonetheless dominate.

A weak state enters the institution if its expected bargaining payoff exceeds the entry cost $c > 0$.⁸ Because unanimity gives the hegemon a higher payoff conditional on participation (as shown in Section 7 below), it leaves less surplus for weak states, making entry harder: $\tau(U) \geq \tau(M)$.

When the institution does not form, the hegemon receives $\alpha V_e(\mu)$ from its bilateral alternative. Since this outside-option payoff is affine in μ and common to both rules, the institutional comparison depends only on the gain from bargaining relative to this baseline.

Definition 2 (Net gain function). *The hegemon's net gain from the institution under rule R at posterior μ is*

$$v(\mu, R) \equiv \begin{cases} E_\theta[V_H^{R1}(\theta, \mu, R)] - \alpha V_e(\mu) & \text{if weak states enter,} \\ 0 & \text{otherwise.} \end{cases}$$

Under majority, $v(\mu, M)$ is affine above the entry threshold. Under unanimity, the screening jump at μ_s^{R1} creates a discontinuity that gives the hegemon a higher net gain at every belief where entry

⁸The entry cost c may represent fixed costs of negotiating accession, per-round costs of preparing delegations, or political costs of committing to multilateral disciplines. In established organizations, c is best interpreted as the cost of *substantive engagement*: allocating scarce diplomatic and analytical resources to a particular negotiating round. Since $V(\theta)$ is normalized independently of N , per-capita payoffs scale as $O(1/N)$. For large- N applications, c should be read as a per-capita cost: writing $c = \tilde{c}/N$, the entry condition becomes $N \cdot V_W \geq \tilde{c}$, where $N \cdot V_W$ is $O(1)$. All equilibrium cutoffs (μ_s^{R1} , μ_s^{R2} , α^*) are homogeneous of degree zero in the surplus and thus independent of this normalization.

occurs (Theorem~1).

7 Institutional Choice

This section establishes the paper’s main result: conditions under which the hegemon prefers unanimity.

7.1 Conditional payoff comparison

Before turning to the full institutional comparison, I establish the paper’s central result: conditional on the institution forming, unanimity gives the hegemon a strictly higher expected continuation payoff than majority for all interior beliefs.

Define the *conditional dominance threshold*

$$\alpha^*(N, \beta) \equiv \frac{\beta(q-1)}{N(N-1) - \beta(N^2 - N - q + 1)}, \quad q = \lfloor N/2 \rfloor + 1. \quad (4)$$

The parameter α measures the strength of the hegemon’s bilateral outside option relative to multilateral cooperation. The threshold α^* marks the point at which the cost of buying unanimous agreement— $N-1$ votes, versus $q-1$ under majority—outweighs the screening advantage. When $\alpha < \alpha^*$, the screening rent from unanimity exceeds the higher vote-buying cost at every belief; when $\alpha \geq \alpha^*$, this fails near $\mu = 1$, where screening rents vanish and the cost difference is decisive.

Theorem 1 (Conditional payoff dominance of unanimity). *The condition $\alpha < \alpha^*(N, \beta)$ holds if and only if, for every $\mu \in (0, 1]$,*

$$E_\theta[V_H^{R1}(\theta, \mu, U)] > E_\theta[V_H^{R1}(\theta, \mu, M)].$$

Proof. See Appendix B.5. □

The condition $\alpha < \alpha^*(N, \beta)$ is pinned down by the no-uncertainty endpoint $\mu = 1$. At that belief, screening rents vanish: the hegemon is known to be the high type, so unanimity can dominate

majority only if its bargaining advantage exceeds the cost of buying all votes. The appendix shows that this is the tightest case: once $D(1) > 0$, the conditional payoff advantage of unanimity, $D(\mu) \equiv E[V_H(\mu, U)] - E[V_H(\mu, M)]$, remains strictly positive for all beliefs $\mu \in (0, 1]$. The intuition has two parts. When the hegemon proposes, unanimity depresses weak states' continuation values (because rejection leads to a game where the hegemon is pivotal), allowing the hegemon to buy agreement more cheaply. When a weak state proposes, unanimity forces the proposer to secure the hegemon's vote, generating the screening rent identified in Proposition 3. Both forces favor the hegemon.

Because unanimity's vote-buying cost grows with N while majority's grows more slowly, α^* decreases with organization size: for $\beta = 0.9$, $\alpha^* \approx 0.47$ when $N = 5$ but falls to ≈ 0.03 when $N = 164$. The condition is easy to satisfy in small institutions but demanding in large ones. For a WTO-sized organization, conditional dominance requires that the hegemon's bilateral alternatives capture only a small fraction of the multilateral surplus. This does not eliminate the mechanism—the screening jump persists regardless of α —but it restricts the domain over which unanimity dominates at *every* belief. Section 8.2 discusses what happens outside this domain.

Remark 1 (Robustness beyond α^*). *When $\alpha \geq \alpha^*$, the conditional advantage of unanimity fails only near $\mu = 1$. On the conservative branch ($\mu > \mu_s^{R1}$), the payoff difference decomposes as*

$$D(\mu) = \underbrace{D(1)}_{\text{without info}} + (1 - \mu) \cdot \underbrace{\Gamma}_{\text{value of uncertainty}},$$

where $C_{\text{buy}} \equiv \beta(q - 1)(1 - \alpha)$ is the vote-buying cost under majority, $C_{\text{out}} \equiv N(N - 1)\alpha$ is the aggregate outside-option cost, $D(1) = r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]/N^2 \leq 0$ is the payoff comparison when the hegemon's type is known, and $\Gamma = (r - 1)[C_{\text{out}} - C_{\text{buy}} + (N - 1)\beta]/N^2 > 0$ is the marginal value of uncertainty for unanimity. The belief threshold above which majority dominates is $\bar{\mu} = 1 - |D(1)|/\Gamma$. Numerically, $\bar{\mu}$ is remarkably stable: for $N = 164$ and $\beta = 0.9$, increasing α from 0.05 to 0.49 (an 18-fold increase beyond α^*) only lowers $\bar{\mu}$ from 0.87 to 0.71 when $r = 1.5$. The mechanism fails only near certainty about the magnitude of the gains from cooperation—precisely where private information has least value (Figure 5 in Section 8.2 maps the full (α, μ) frontier).

7.2 Formation sets and institutional comparison

Let $\mathcal{F}_R \equiv \{p \in (0, 1] : V_W^{R1}(p, R) \geq c\}$ denote the *formation set* under rule R : the set of priors at which the institution forms. Since weak states' aggregate payoff is weakly lower under unanimity whenever the hegemon's payoff is higher (Theorem~1), the unanimity entry constraint is harder to satisfy; hence $\mathcal{F}_U \subseteq \mathcal{F}_M$ (Appendix B.6). The institutional comparison is therefore determined by the entry margin: unanimity dominates when the institution forms, but majority can dominate when only it can induce entry.

Corollary 1 (Unanimity dominance on the formation set). *Suppose $\alpha < \alpha^*(N, \beta)$. For every prior $p \in \mathcal{F}_U$,*

$$V_H(U, p) > V_H(M, p).$$

Moreover, $\mathcal{F}_U \subseteq \mathcal{F}_M$.

Proof. See Appendix B.6. □

Remark 2 (Weak states prefer majority). *Under the conditions of Theorem 1, $V_W(\mu, M) > V_W(\mu, U)$ for every $\mu \in (0, 1]$, and $\mathcal{F}_U \subseteq \mathcal{F}_M$. Unanimity redistributes from weak states to the hegemon; it is never Pareto-improving conditional on entry.*

Why, then, would weak states participate under unanimity? Because the hegemon chooses the voting rule. A weak state that refuses to enter under unanimity does not get majority instead—it gets no institution at all, forfeiting whatever surplus the organization would generate. Since weak states are ex ante identical and decide simultaneously, the symmetric equilibrium is all-or-nothing: either all enter or none does (Section 3). There is no scope for individual holdout to force a rule change. Conditional on the hegemon's choice of unanimity, entry is individually rational whenever $V_W(p, U) \geq c$. The irony is that the veto, which guarantees each weak state inclusion in every agreement, is precisely what creates the screening problem that benefits the hegemon: by guaranteeing inclusion, unanimity forces weak proposers to make offers without knowing the hegemon's type, generating the overpayment that drives the conditional payoff advantage.

The following result completes the institutional comparison by classifying all priors.

Proposition 4 (Institutional classification). *Suppose $\alpha < \alpha^*(N, \beta)$. The hegemon's preferred rule*

is characterized as follows:

$$\begin{aligned}
U \succ M & \text{ if and only if } p \in \mathcal{F}_U, \\
M \succ U & \text{ if and only if } p \in \mathcal{F}_M \setminus \mathcal{F}_U, \\
U \sim M & \text{ if and only if } p \notin \mathcal{F}_M.
\end{aligned}$$

Since $\mathcal{F}_U \subseteq \mathcal{F}_M$, majority's only advantage is that it can support institutional entry at priors for which unanimity cannot.

Proof. See Appendix B.8. □

When the prior is such that weak states find it individually rational to enter under unanimity, the hegemon strictly prefers it (Corollary~1). When only majority can sustain entry, majority dominates—not through better bargaining terms, but through wider institutional viability. When neither rule can induce entry, the voting rule is irrelevant. When entry costs are low enough that $\mathcal{F}_U = (0, 1]$, unanimity dominates at every prior.

Remark 3 (Information design). *The model also clarifies why information design would matter differently under the two voting rules. Under majority rule, the hegemon's continuation payoff is affine in weak states' posterior belief once entry is feasible. Splitting beliefs therefore does not create additional value beyond the payoff implied by the prior. Under unanimity, by contrast, weak states must include the hegemon in any agreement. This creates a screening discontinuity at the posterior threshold at which entry becomes feasible. As in standard Bayesian persuasion [Kamenica2011], such nonconcavities are precisely where information design can matter. Thus, although persuasion is not necessary for the paper's main result, the model identifies why consensus institutions are more hospitable to informational leverage than majority institutions.*

Example 2 (The full institutional comparison). *Continuing with the parameters from Example 1 ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$), suppose entry costs $c = 0.14$. Under majority, the institution forms at essentially all beliefs ($\tau(M) \approx 0$), so $\mathcal{F}_M \approx (0, 1]$. Under unanimity, the screening mechanism reduces weak states' surplus, so the institution forms only for $p \gtrsim 0.33$: $\mathcal{F}_U \approx [0.33, 1]$.*

For $p \in \mathcal{F}_U$ (above ≈ 0.33), unanimity strictly dominates: at $p = 0.50$, the hegemon's payoff under unanimity exceeds majority by 25%. For $p \in \mathcal{F}_M \setminus \mathcal{F}_U$ (below ≈ 0.33), majority dominates because only it can sustain entry. The advantage of majority at pessimistic priors is entirely through the entry channel: conditional on participation, unanimity always gives the hegemon more (Theorem 1).

7.3 Numerical characterization

Figure 4 maps the institutional classification directly. For each combination of prior p and entry cost c , the figure shows which rule the hegemon prefers: unanimity (blue), majority through the entry channel (orange), or neither (gray). The unanimity-dominance region is always nested inside the majority-entry region, confirming that $\mathcal{F}_U \subseteq \mathcal{F}_M$. The unanimity-dominance region expands with r (larger screening rents from greater type dispersion) and β (more patient bargaining strengthens the off-path threat that sustains screening), and contracts with α (stronger outside options raise the cost of buying unanimous agreement) and c (higher entry costs shrink $\mathcal{F}_U$ faster than $\mathcal{F}_M$, widening majority's scope advantage). The figure also illustrates why entry under unanimity need not be monotone in the prior. The same screening rent that gives the hegemon a conditional advantage under unanimity lowers weak-state payoffs at the cutoff, creating a local failure of entry under unanimity even when entry is feasible at lower and higher priors.

8 Discussion

8.1 Illustration: The GATT/WTO

The WTO illustrates the interaction between consensus and informational asymmetry. The multi-lateral trading system operates by consensus, even though formal voting provisions exist (Steinberg 2002). The United States and other major trading powers could plausibly press for majority rule, yet consensus persists.

The model's informational asymmetry assumption finds support in the trade context. Evaluating the effects of tariff schedules, anticipating distributive consequences, and assessing long-term implications of rule changes require sustained investment in analytical infrastructure. Major trading powers

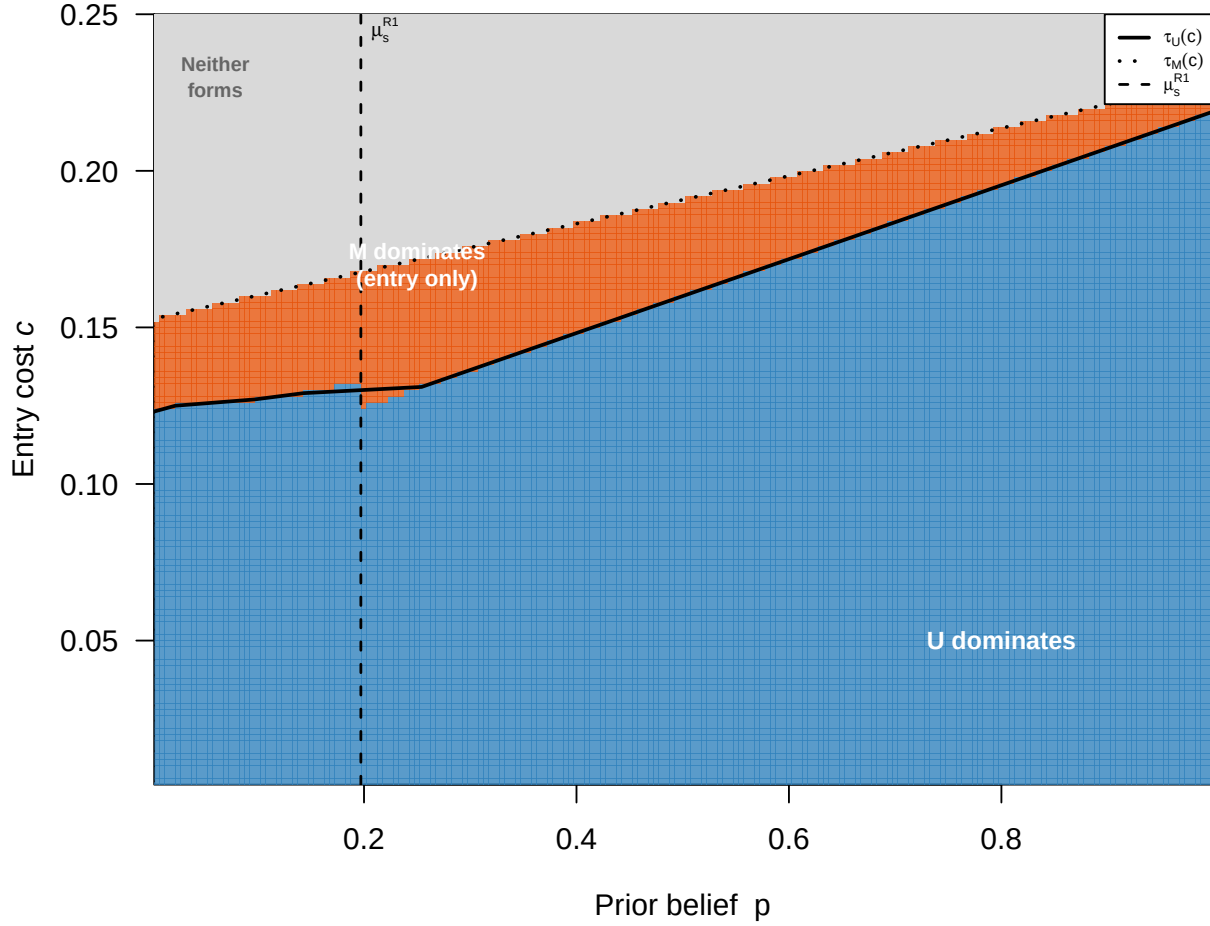


Figure 4: Institutional classification in the (p, c) plane ($N = 5, r = 1.5, \alpha = 0.3, \beta = 0.9$). Blue indicates pairs for which unanimity sustains entry and therefore dominates majority conditional on participation. Orange indicates pairs for which majority sustains entry but unanimity does not, so majority dominates through the entry channel. Gray indicates pairs for which neither rule sustains entry. The vertical dashed line marks the screening cutoff μ_s^{R1} . At this cutoff, weak proposers switch from aggressive to conservative offers under unanimity, generating an upward jump in the hegemon's payoff and, by budget balance, a downward jump in weak-state payoffs. This can make the unanimity formation set disconnected.

maintain large permanent delegations with deep sectoral expertise; many developing country delegations operate with far smaller teams, particularly during complex technical phases of negotiations (Jawara and Kwa 2003; Jones, Deere-Birkbeck, and Woods 2010; Blackhurst, Lyakurwa, and Oyejide 2000). This capacity asymmetry means that weaker states often face genuine uncertainty about the value of proposed agreements.

In practice, major trading powers exercise informal agenda influence through the Green Room process, control over drafting, and technical expertise. The model's baseline uses symmetric proposal rights to isolate the voting-rule comparison, but the qualitative mechanism—screening under unanimity, no screening under majority—does not depend on equal recognition probabilities. Monopoly agenda control would eliminate participation (as argued in Section 8.2), so the relevant question is whether moderate informal influence coexists with the screening channel. The persistence of consensus is consistent with the coexistence of informal influence and formal equality.

The model's mechanism provides an alternative account of a pattern not fully explained to date: powerful states in the WTO appear to benefit from consensus despite lacking formal agenda control. The mechanism is not merely that informal practices survive consensus, but that consensus creates the strategic environment—pivotal inclusion under uncertainty—in which informational power becomes most valuable. The model does not claim that the GATT/WTO was designed for this reason, or that all aspects of WTO politics reduce to this mechanism. It shows that consensus and informational asymmetry interact in a way that can benefit powerful states, offering one explanation for why hegemonic preference for consensus is not paradoxical.

The model generates a discriminating empirical prediction through the selection channel. Because unanimity is sustainable only when the hegemon's informational advantage generates screening rents, and screening rents require that weaker states face genuine uncertainty about the value of cooperation, the model predicts a negative association between consensus rules and transparent distributive stakes in the cross-section of surviving international organizations. When the gains from cooperation are easily calculable—as in financial institutions where capital contributions, loan terms, and voting shares create a measurable pie—weaker states can evaluate proposals without informational disadvantage, screening rents vanish, and unanimity loses its conditional payoff advantage. Such organizations should either not adopt unanimity, adopt it and subsequently migrate to

majority or weighted voting, or fail to form under unanimity at all. When the value of cooperation is harder to assess *ex ante*—as in regulatory coordination or standard-setting bodies where the consequences of agreements depend on implementation, market responses, and enforcement trajectories that are difficult to forecast—the informational asymmetry that drives the screening mechanism is most relevant, and consensus should be both preferred by the hegemon and sustainable.⁹

This prediction distinguishes the informational mechanism from the main alternatives. Legitimacy accounts predict consensus where binding obligations require broad-based assent, without conditioning on whether the distributive consequences of cooperation are transparent or opaque. Self-binding accounts predict consensus where commitment problems are most severe, regardless of informational structure. Informal power accounts predict that powerful states exercise influence irrespective of the formal voting rule, and therefore cannot explain why consensus is systematically absent in financial organizations where powerful states have the strongest informal leverage. The informational mechanism, by contrast, predicts that the observable distribution of voting rules is shaped by a selection process: organizations that combine unanimity with transparent distributive stakes are unstable. Existing data on the voting rules of international organizations (Gould 2022) are consistent with this pattern—financial institutions overwhelmingly use weighted majority voting, while organizations concerned with regulatory coordination are the most frequent adopters of consensus—but a formal test of the interaction between voting rules and informational structure remains an open question. The model further predicts that as weaker states develop independent capacity to evaluate cooperation value—through investment in policy research, participation in technical committees, or access to third-party assessments—the screening advantage under consensus should diminish, narrowing the set of issue areas where consensus is sustainable. The proliferation of preferential trade agreements, which strengthens the hegemon’s bilateral alternatives, should independently weaken the preference for consensus by raising α toward α^* (Bhagwati 2008). The mechanism identified here is complementary to existing accounts of institutional design, isolating a distributive-informational channel within the rationalist framework common to all of them (cf. Fearon 1995).

⁹Appendix C shows that with $K > 2$ types, unanimity generates $K - 1$ screening boundaries while majority remains linear. The screening rent is strictly positive for any continuous type distribution (Appendix C.4), and for the uniform distribution the dominance threshold is weakly larger than in the binary case, suggesting that the binary model is the most conservative specification for the mechanism.

8.2 Scope

Why symmetric proposal rights? Under either voting rule, if the hegemon were the exclusive proposer and weak states had no outside option from bargaining ($d_W = 0$), weak states' expected bargaining payoff would be zero. No weak state would pay an entry cost $c > 0$, and the institution would never form. Monopoly proposal power is self-defeating when participation is voluntary. The relevant institutional choice is therefore between voting rules, holding proposal rights symmetric.

Why not compare consensus with hegemonic agenda control? Because agenda control combined with any participation constraint eliminates the institution itself. The puzzle is not why a hegemon does not impose dictatorship, but why it would choose a rule that makes its own approval necessary when majority would allow others to bypass it.

Why simultaneous entry? The all-or-nothing entry technology isolates the screening mechanism by suppressing dynamics that would arise under sequential accession—such as free-riding, learning from early joiners, or collective action problems among weak states. With sequential entry, multiple mechanisms would operate in opposing directions, and the net effect on the hegemon's institutional preference would depend on their relative magnitudes. The present model focuses on the screening channel alone; characterizing its interaction with sequential entry dynamics is left open.

Why binary types? The qualitative result—unanimity creates screening that majority eliminates—holds for any finite type space (Appendix C). The quantitative result—unanimity conditionally dominates at *every* belief (Theorem 1)—is strongest with binary types. With richer type spaces, unanimity generates $K - 1$ screening boundaries, but the parametric region of global conditional dominance may narrow. Binary types isolate the mechanism most transparently; the model is the most favorable case for the result.

When consensus favors the hegemon. The mechanism requires that: (i) informational asymmetry is relevant (H knows θ , W does not); (ii) the hegemon has a valuable outside option ($\alpha > 0$); (iii) entry is costly and belief-dependent; (iv) the voting rule makes H pivotal.

When majority dominates. The biconditional in Theorem 1 implies that $\alpha < \alpha^*$ is the largest parametric region where majority can never dominate through a higher conditional-on-entry payoff. Any advantage of majority must come through the entry margin: $\tau(U) > \tau(M)$. This is

more likely when entry costs are high (creating a large gap where the institution forms only under majority), when α approaches α^* (weakening the conditional dominance), or when information is approximately public. For $\alpha \geq \alpha^*$, majority can also dominate through conditional bargaining payoffs near $\mu = 1$, because unanimity's cost of buying $N - 1$ votes outweighs the screening advantage.

The threshold α^* decreases with N because unanimity requires buying $N - 1$ votes, while majority requires only $q - 1$. For large organizations, α^* becomes small: the conditional dominance result applies only when the hegemon's bilateral alternative is modest relative to multilateral cooperation. This does not eliminate the mechanism: even when Theorem 1 fails globally, the screening jump at μ_s^{R1} persists, and unanimity can still dominate at beliefs where the screening rent outweighs the vote-buying cost. The condition restricts when unanimity dominates *everywhere*, not when it dominates *somewhere*. Figure 5 illustrates: comparing the $N = 30$ and $N = 164$ panels, α^* (dashed line) shifts left, but the blue region—where unanimity dominates—remains extensive.

The biconditional in Theorem 1 implies that this boundary is sharp. At $\alpha = \alpha^*$, the dominance difference $D(\mu)$ equals zero exactly at $\mu = 1$; for $\alpha > \alpha^*$, majority strictly dominates near $\mu = 1$ (where the vote-buying cost is decisive), while unanimity can still dominate at lower beliefs. The paper's main results are stated for $\alpha < \alpha^*$, which ensures a clean separation between the conditional and entry channels.

Figure 5 maps the conditional dominance region in the (α, μ) plane. For each parameter combination, the blue region indicates beliefs at which unanimity gives the hegemon a higher continuation payoff; the red region indicates beliefs at which majority dominates. The solid curve traces the frontier $\bar{\mu}(\alpha)$ from Remark 1. Two patterns stand out. First, the frontier is remarkably flat: increasing α well beyond α^* shifts $\bar{\mu}$ only modestly downward, confirming that the screening advantage is robust across most of the belief space. Second, the red region is always concentrated near $\mu = 1$ —near certainty about the magnitude of the gains from cooperation—where informational asymmetry is irrelevant and the cost of buying unanimous agreement dominates.

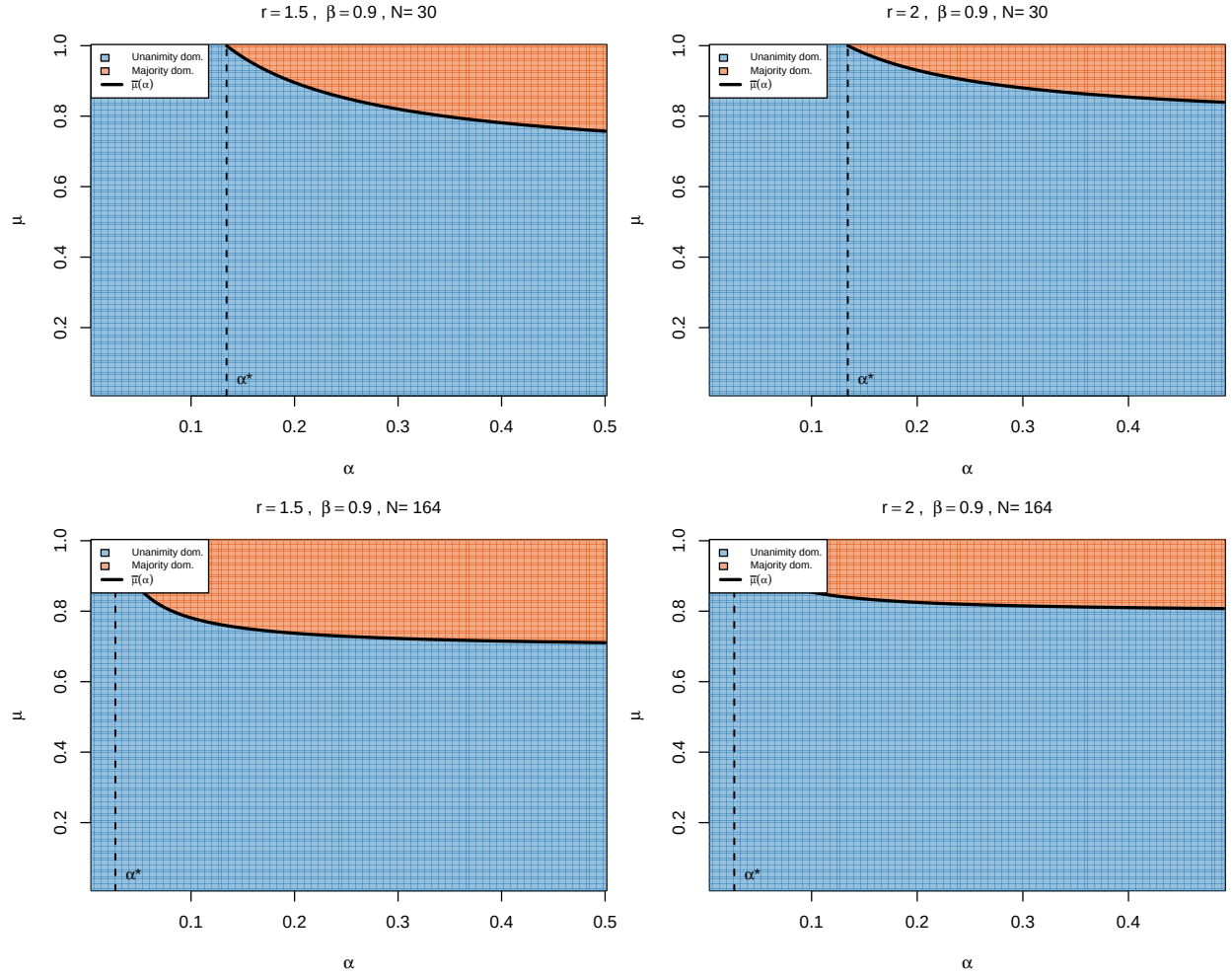


Figure 5: Conditional payoff dominance in the (α, μ) plane. Blue: unanimity dominates ($D(\mu) > 0$). Red: majority dominates ($D(\mu) < 0$). Solid curve: belief threshold $\bar{\mu}(\alpha)$ above which majority dominates (Remark~1). Dashed line: α^* . Left column: $r = 1.5$. Right column: $r = 2$. Top row: $N = 30$. Bottom row: $N = 164$. All panels: $\beta = 0.9$.

9 Conclusion

This paper has identified conditions under which consensus is a rational choice for a hegemon. The key is to distinguish between two modes of exercising power under different voting rules. Majority rule with symmetric proposals lets weak states bypass the hegemon, eliminating the screening problem. Unanimity forces weak states to include the hegemon under uncertainty, creating a screening cutoff that gives the hegemon a conditional payoff advantage at every level of uncertainty. The institutional comparison has a clean structure: unanimity dominates wherever it can sustain entry, and majority's only advantage is wider institutional viability.

The argument does not imply that all consensus institutions are designed for this reason, or that informal power disappears under consensus. It shows something more specific: under identifiable conditions on informational asymmetry and the prospects of cooperation, unanimity becomes the preferred institutional arrangement for a hegemon. Consensus may be less a concession by power than a way of exercising power through a different channel—the channel of information.

The main results use a binary type space. With $K > 2$ types, Appendix C shows that unanimity generates $K - 1$ screening boundaries while majority remains linear. The qualitative mechanism—unanimity creates screening, screening creates a conditional payoff advantage—extends to arbitrary type distributions (Appendix C.4 proves that the screening rent is strictly positive for any continuous distribution on $[1, r]$). For the uniform distribution, the conditional payoff comparison admits a closed-form solution, and the dominance threshold α_{cont}^* is weakly larger than the binary threshold α^* , suggesting that the binary model is the most conservative case for the mechanism rather than the most favorable one. The intuition is that continuous types generate richer overpayment: with many types below the screening cutoff, the aggregate rent from pooling increases. A general characterization of α^* for arbitrary distributions remains open, but the available evidence—the qualitative result for general distributions and the quantitative result for the uniform—points toward robustness.

Several extensions remain. First, if weaker states can invest in independent analytical capacity, informational power erodes endogenously over time. Second, heterogeneous outside options across weak states (middle powers) would enrich the screening dynamics. Third, repeated play within an established consensus institution could generate learning that narrows the informational gap. These directions point toward a richer theory of how informational power rises and declines within

institutional frameworks.

Appendix A: Bargaining Derivations

Notation summary

Symbol	Meaning	Defined in
N	Number of players (1 hegemon + $N - 1$ weak states)	Def. 1
$\theta \in \{0, 1\}$	State of the world	Def. 1
p	Prior $\Pr(\theta = 1)$	Def. 1
μ	Belief $\Pr(\theta = 1)$; equals prior p at entry, updated during bargaining	Sec. 3
$V(\theta)$	Value of cooperation: $V(0) = 1$, $V(1) = r$	Def. 1
$r > 1$	High-type value	Def. 1
$\alpha \in (0, 1/r)$	Outside-option share: $d_H = \alpha V(\theta)$	Def. 1
$\beta \in (0, 1)$	Common discount factor	Def. 1
$c > 0$	Entry cost per weak state	Def. 1
$V_e(\mu)$	Expected cooperation value $1 + \mu(r - 1)$	Sec. 3
x	Shorthand $(N - 1)\alpha r$	Sec. 3
q	Majority quota $\lfloor N/2 \rfloor + 1$	Def. 1
$R \in \{M, U\}$	Voting rule (majority / unanimity)	Def. 1
μ_s^{R2}	R2 screening cutoff	App. A.2
μ_s^{R1}	R1 screening cutoff	Prop. 2
ϕ	Auxiliary for R1 cutoff: $[rN - \beta(N - 1 + r)]/[\beta(r - 1)]$	Prop. 2
$\tau(R)$	Entry threshold under rule R	Sec. 6
$v(\mu, R)$	Hegemon's net gain from institution: $E[V_H^{R1}] - \alpha V_e(\mu)$ if entry, 0 otherwise	Sec. 6
$\bar{\alpha}$	Regime threshold for α -independent R1 cutoff	Prop. 2
α^*	Threshold for conditional dominance (Thm. 1): necessary and sufficient	Eq. (4)
$\omega(\mu)$	R2 continuation shorthand: $(N - 2)\beta V_W^{R2}(\mu)$	App. A.3
κ_M	Majority weak-state coefficient: $(1 - \alpha)[N(N - 1) + \beta(q - 1)]/(N^2(N - 1))$	App. A.7
$D(\mu)$	Conditional payoff difference $V_H^{R1}(\mu, U) - V_H^{R1}(\mu, M)$	Thm. 1
Γ	Marginal value of uncertainty for unanimity	Remark 1
$\bar{\mu}$	Belief threshold above which majority dominates: $1 - D(1) /\Gamma$	Remark 1
$C_{\text{buy}}, C_{\text{out}}$	Vote-buying cost, outside-option cost (proof)	App. B.5
λ_M	Majority payoff coefficient	App. B.5
$\mathcal{F}_R$	Formation set under rule R : $\{p : V_W^{R1}(p, R) \geq c\}$	Sec. 7

A.1 Majority: R2 and R1

Under majority, weak proposers form a winning coalition from other weak states, excluding H . The hegemon captures $\alpha V(\theta)$ from its bilateral alternative. H 's payoff is type-specific and does not depend on W 's beliefs.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N-1)\alpha]}{N} \quad (5)$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N} \quad (6)$$

R1 values follow by backward induction with the same structure (no screening, linear).

A.2 Unanimity: R2

When H proposes: offers 0 to all W 's, keeps $V(\theta)$.

When W proposes: the screening subgame has cutoff $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$. Below it, W plays aggressive ($y_H = \alpha$); above, conservative ($y_H = \alpha r$).

The hegemon's R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1 + (N-1)\alpha]}{N} \quad (\text{constant in } \mu), \quad (7)$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1 + (N-1)\alpha}{N}, \quad (8)$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1 + (N-1)\alpha r}{N} \quad (\text{overpaid}). \quad (9)$$

Weak states' R2 continuation values under unanimity:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1-\mu)(1-\alpha)}{N} \quad (10)$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N} \quad (11)$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 screening derivation

The proposing W compares:

$$F_1^{con}(\mu) = V_e(\mu) - \frac{\beta(r+x)}{N} - \omega(\mu) \quad (12)$$

$$F_1^{agg}(\mu) = (1-\mu) \left[1 - \frac{\beta(1+x)}{N} - \omega(\mu) \right] + \frac{\mu\beta r(1-\alpha)}{N} \quad (13)$$

where $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$.

The difference $\Delta_1(\mu) = F_1^{agg} - F_1^{con}$ is piecewise quadratic. On the high branch ($\mu > \mu_s^{R2}$), the α -dependent terms cancel, yielding the quadratic:

$$(N-2)\beta(r-1)\mu^2 + [\beta(N-1+r) - rN]\mu + \beta(r-1) = 0.$$

Dividing by $\beta(r-1)$:

$$(N-2)\mu^2 - \phi\mu + 1 = 0, \quad \phi = \frac{rN - \beta(N-1+r)}{\beta(r-1)}.$$

The smaller root is μ_s^{R1} (equation (1)).

A.4 Jump magnitude

At the R1 cutoff, type $\theta = 0$ receives $\beta(1+x)/N$ (aggressive) vs $\beta(r+x)/N$ (conservative) when W proposes. The difference $\beta(r-1)/N$, weighted by $(N-1)/N$ (prob W proposes) and $(1 - \mu_s^{R1})$ (prob $\theta = 0$), gives the jump in equation (3).

A.5 Case selection

The R1 screening cutoff lies on the high R2 branch ($\mu_s^{R1} > \mu_s^{R2}$, α -independent) when $\Delta_1(\mu_s^{R2}) > 0$. On the high branch, Δ_1 is a convex quadratic with $\Delta_1(0) > 0$ and $\Delta_1(1) = rN(\beta-1) < 0$, so $\Delta_1(\mu) > 0$ for all $\mu < \mu_s^{R1}$. Hence $\Delta_1(\mu_s^{R2}) > 0$ if and only if $\mu_s^{R2} < \mu_s^{R1}$. Substituting

$\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$ into $\mu_s^{R2} < \mu_s^{R1}$ and solving for α :

$$\frac{\alpha(r-1)}{r-\alpha} < \mu_s^{R1} \iff \alpha < \frac{\mu_s^{R1} \cdot r}{r-1+\mu_s^{R1}} = \bar{\alpha}(r, \beta, N),$$

which defines the threshold in equation (2).

When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch, where $V_W^{R2}(\mu) = (1-\mu)(1-\alpha)/N$ and Δ_1 depends on α through $\omega(\mu)$. The cutoff still exists and is unique: $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ hold regardless of branch, so a root exists by the intermediate value theorem; since Δ_1 is quadratic on $(0, \mu_s^{R2})$ (at most two roots) with opposite-sign endpoints, there is exactly one root. On the high branch, $\Delta_1 \leq 0$, so no additional roots exist. The jump at μ_s^{R1} remains positive. The proof of Theorem~1 (Appendix~B.5) covers both regimes explicitly: the payoff decomposition $D = D_{\text{base}} + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2} + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}$ is additive regardless of the relative ordering of μ_s^{R1} and μ_s^{R2} , and the endpoint argument establishes $D(\mu) > 0$ on all branches in both cases.

A.6 Budget checks

Under majority, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds exactly: $V_H[1 + (N-1)\alpha]/N + (N-1)(1-\alpha)V_e(\mu)/N = V_e(\mu) \cdot [1 + (N-1)\alpha + (N-1)(1-\alpha)]/N = V_e(\mu)$. Under unanimity, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds on the conservative R1 branch, where all deals pass in R1. On the aggressive R1 branch, surplus destruction from discounting when $\theta = 1$ rejects in R1 and the game proceeds to R2 yields $E[V_H] + (N-1)V_W = V_e(\mu) - \frac{(N-1)\mu r(1-\beta)}{N}$, so the identity holds with inequality: $E[V_H] + (N-1)V_W \leq V_e(\mu)$.

A.7 Entry thresholds

Under majority, $V_W^{R1}(\mu, M) = \kappa_M V_e(\mu)$ where $\kappa_M = (1-\alpha)[N(N-1) + \beta(q-1)]/(N^2(N-1))$. The entry threshold is $\tau(M) = \max\{0, (c/\kappa_M - 1)/(r-1)\}$.

Under unanimity, V_W^{R1} is piecewise. On the conservative R1 branch ($\mu > \mu_s^{R1}$, which also implies $\mu > \mu_s^{R2}$ when $\alpha < \bar{\alpha}$):

$$V_W^{R1}(\mu, U) = \frac{(N+\beta)V_e(\mu) - \beta r(1+N\alpha)}{N^2}.$$

This is affine in μ . Setting $V_W^{R1} = c$ and solving:

$$\tau_U^{con} = \frac{cN^2 - [N + \beta - \beta r(1 + N\alpha)]}{(N + \beta)(r - 1)}. \quad (14)$$

On the aggressive R1 branch ($\mu < \mu_s^{R1}$), the proposer surplus F_1^{agg} is quadratic in μ (the product of $(1 - \mu)$ with the affine terms in $\omega(\mu)$ generates a second-order term). However, the full weak-state expected payoff V_W^{R1} —which averages over proposer and non-proposer roles—simplifies to a piecewise affine expression on each R2 sub-branch, because the quadratic terms in the proposer and non-proposer components cancel (see the proof of Lemma 1 in Appendix B.7). The entry threshold on each sub-branch therefore solves a linear equation.¹⁰

The numerical results in the paper compute $\tau(U)$ by pointwise evaluation of V_W^{R1} .

Appendix B: Proofs

B.1 Proof of Proposition 1 (Majority produces no screening)

Under majority rule, W 's proposal passes with $q - 1$ votes from other weak states. H 's vote is not required. W excludes H from the winning coalition; H captures $\alpha V(\theta)$ from its bilateral alternative regardless of μ . There is no screening and no jump. All continuation values are affine in $V_e(\mu)$, hence affine in μ . $\square$

B.2 Proof of Proposition 2 (R1 cutoff existence and uniqueness)

See Appendix A.3 for the derivation. *Existence:* $\Delta_1(0) = \beta(r - 1)/N > 0$ and $\Delta_1(1) = rN(\beta - 1) < 0$; by continuity, a root exists in $(0, 1)$.

Uniqueness and closed form under $\alpha < \bar{\alpha}$: When $\alpha < \bar{\alpha}(r, \beta, N)$, the root lies on the high R2 branch ($\mu > \mu_s^{R2}$), where Δ_1 is a convex quadratic in μ with coefficients independent of α (Appendix A.3). Since $\Delta_1(\mu_s^{R2}) > 0$ (Appendix A.5) and $\Delta_1(1) < 0$, there is exactly one root

¹⁰Because V_W^{R1} has a downward jump at μ_s^{R1} (opposite to V_H 's upward jump), the formation set $\{\mu : V_W^{R1} \geq c\}$ can be disconnected for intermediate values of c : entry may occur both below μ_s^{R1} (aggressive regime) and above some $\mu_2 > \mu_s^{R1}$ (conservative regime), with a gap around the cutoff. The affine formula (14) gives the entry threshold in the conservative regime and is the relevant quantity when $c > \max_{\mu \leq \mu_s^{R1}} V_W^{R1}(\mu, \text{agg})$.

in $(\mu_s^{R2}, 1)$, given by equation (1). Below μ_s^{R2} , Δ_1 is a concave quadratic in μ : the coefficient of μ^2 is $-(N-2)\beta(1-\alpha)/N < 0$, arising from the product $(1-\mu) \cdot \omega(\mu)$ in F_1^{agg} where $\omega(\mu) = (N-2)\beta(1-\mu)(1-\alpha)/N$ is linear in μ . Since $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(\mu_s^{R2}) > 0$, and a concave function that is positive at both endpoints of an interval is positive throughout, $\Delta_1(\mu) > 0$ for all $\mu \in [0, \mu_s^{R2}]$. No additional roots exist.

Uniqueness under $\alpha \geq \bar{\alpha}$: The root falls on the low R2 branch, where Δ_1 is a quadratic in μ that depends on α . On this branch, $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ (since $\mu_s^{R2} \geq \mu_s^{R1}$ on the high branch implies $\Delta_1^{\text{high}}(\mu_s^{R2}) \leq 0$, and continuity gives $\Delta_1(\mu_s^{R2}) \leq 0$). Since Δ_1 is quadratic on $(0, \mu_s^{R2})$, it has at most two roots; combined with $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$, there is exactly one root in $(0, \mu_s^{R2}]$. On the high branch, $\Delta_1 \leq 0$ for $\mu \in [\mu_s^{R2}, 1]$, so no additional roots exist there. $\square$

B.3 Proof of Proposition 3 (Jump)

Direct computation from the difference in H's R1 payoff between the conservative and aggressive branches. See Appendix A.4. $\square$

B.5a Derivation of the payoff decomposition

This appendix derives the decomposition $D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu)$ from the primitive payoff formulas in Appendices A.1–A.3.

Step 0: Primitive payoff components. The first task is to write down the hegemon's R1 payoff from its primitive components. Because proposals are random, the payoff splits naturally by proposer identity, and this separation will prove essential: each component turns out to depend on a different screening regime, which is ultimately why the two corrections are additive.

The hegemon's expected R1 payoff $E[V_H^{R1}(\mu, U)]$ has two components, corresponding to the identity of the R1 proposer.

H proposes (probability $1/N$). The hegemon offers each weak state $\beta V_W^{R2}(\mu)$ (its discounted R2

continuation) and keeps the remainder. The expected payoff across types is:

$$\Pi_H^{\text{prop}}(\mu) = \frac{V_e(\mu) - (N-1)\beta V_W^{R2}(\mu)}{N}. \quad (15)$$

W proposes (probability $(N-1)/N$). The proposing weak state makes an offer to H that depends on the R1 screening regime. Under an aggressive R1 offer ($\mu < \mu_s^{R1}$), only $\theta = 0$ accepts; $\theta = 1$ rejects and the game proceeds to R2 with posterior $\mu' = 1$. The R1 offer to H is $\beta V_H^{R2}(\theta=0, \mu'=1)$, and $\theta = 1$'s payoff from rejection is $\beta V_H^{R2}(\theta=1, \mu'=1)$. Since $\mu' = 1 > \mu_s^{R2}$, both R2 values are evaluated in the conservative R2 regime (Appendix A.2, equations (7)–(9)):

$$\beta V_H^{R2}(\theta=0, \mu'=1) = \frac{\beta(1+x)}{N}, \quad (16)$$

$$\beta V_H^{R2}(\theta=1, \mu'=1) = \frac{\beta(r+x)}{N}. \quad (17)$$

Under a conservative R1 offer ($\mu > \mu_s^{R1}$), both types accept. The offer is set at $\beta V_H^{R2}(\theta=1, \mu'=1) = \beta(r+x)/N$ to satisfy the high type's participation constraint. Hence the expected payoff from W 's proposal is:

$$\Pi_H^W(\mu) = \frac{N-1}{N} \begin{cases} \mu \frac{\beta(r+x)}{N} + (1-\mu) \frac{\beta(1+x)}{N} & \text{if } \mu < \mu_s^{R1} \text{ (aggressive R1),} \\ \frac{\beta(r+x)}{N} & \text{if } \mu > \mu_s^{R1} \text{ (conservative R1).} \end{cases} \quad (18)$$

The total hegemon payoff is $E[V_H^{R1}(\mu, U)] = \Pi_H^{\text{prop}}(\mu) + \Pi_H^W(\mu)$.

Two features are important. First, the R1 offers to H in (16)–(17) are evaluated at the off-path posterior $\mu' = 1$ and therefore do not depend on $V_W^{R2}(\mu)$. Second, H 's proposal payoff (15) depends on $V_W^{R2}(\mu)$ but is the same regardless of the R1 regime. These two observations establish that the R1 and R2 corrections are independent.

Step 1: Benchmark payoff (aggressive R1, conservative R2). I now compute the hegemon's total payoff in a benchmark regime—aggressive R1 screening and conservative R2 screening—which serves as the reference point against which the two corrections will be measured. The choice of benchmark is not arbitrary: under this regime, both components of the payoff take their simplest

form, and the corrections for each alternative regime can then be isolated one at a time.

Under aggressive R1 and conservative R2 ($V_W^{R2} = [V_e(\mu) - \alpha r]/N$), the two components are:

$$\Pi_H^{\text{prop}}(\mu)|_{\text{con-R2}} = \frac{V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]/N}{N} = \frac{N V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]}{N^2}, \quad (19)$$

$$\Pi_H^W(\mu)|_{\text{agg-R1}} = \frac{(N-1)\beta}{N} \cdot \frac{V_e(\mu) + x}{N} = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}. \quad (20)$$

Summing:

$$\begin{aligned} E_{\text{bench}}(\mu) &= \frac{N V_e(\mu) - (N-1)\beta V_e(\mu) + (N-1)\beta \alpha r + (N-1)\beta V_e(\mu) + (N-1)\beta x}{N^2} \\ &= \frac{N V_e(\mu) + (N-1)\beta(\alpha r + x)}{N^2} = \frac{N V_e(\mu) + (N-1)\beta N \alpha r}{N^2}, \end{aligned} \quad (21)$$

where the last equality uses $\alpha r + x = \alpha r + (N-1)\alpha r = N\alpha r$.

Step 2: R1 correction δ_{R1} . The next step is to compute the payoff change when the R1 regime switches from aggressive to conservative. Because this switch affects only the R1 offers that W makes to H —and not H 's own proposal, which depends on V_W^{R2} —the correction will appear exclusively in Π_H^W , confirming that it can be treated independently of the R2 regime.

When the R1 regime switches from aggressive to conservative ($\mu > \mu_s^{R1}$), Π_H^{prop} is unchanged (it depends only on V_W^{R2} , not on the R1 regime). The change is entirely in Π_H^W . Under aggressive R1:

$$\Pi_H^W|_{\text{agg}} = \frac{(N-1)\beta}{N^2} [\mu(r+x) + (1-\mu)(1+x)] = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}.$$

Under conservative R1:

$$\Pi_H^W|_{\text{con}} = \frac{(N-1)\beta(r+x)}{N^2}.$$

The correction is:

$$\delta_{R1}(\mu) \equiv \Pi_H^W|_{\text{con}} - \Pi_H^W|_{\text{agg}} = \frac{(N-1)\beta}{N^2} [(r+x) - V_e(\mu) - x] = \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \quad (22)$$

This is non-negative for all $\mu \leq 1$, with equality only at $\mu = 1$. The correction reflects the

overpayment that the low type receives under conservative R1 offers.

Step 3: R2 correction δ_{R2} . I now derive the payoff change when the R2 regime switches from conservative to aggressive. By an argument symmetric to Step 2, this switch affects only H 's proposal payoff Π_H^{prop} —the R1 offers to H are pinned at the off-path posterior $\mu' = 1$, which always lies in the conservative R2 regime regardless of the on-path belief μ .

When the R2 regime switches from conservative to aggressive ($\mu < \mu_s^{R2}$), Π_H^W is unchanged (the R1 offers to H are evaluated at $\mu' = 1$, which is always in the conservative R2 regime). The change is entirely in Π_H^{prop} . Substituting $V_W^{R2, \text{agg}}(\mu) = (1 - \mu)(1 - \alpha)/N$ (Appendix A.2) in place of $V_W^{R2, \text{con}}(\mu) = [V_e(\mu) - \alpha r]/N$:

$$\begin{aligned}\delta_{R2}(\mu) &\equiv \Pi_H^{\text{prop}}|_{\text{agg-R2}} - \Pi_H^{\text{prop}}|_{\text{con-R2}} \\ &= \frac{-(N-1)\beta}{N^2} \left[(1 - \mu)(1 - \alpha) - V_e(\mu) + \alpha r \right] \\ &= \frac{(N-1)\beta}{N^2} \left[V_e(\mu) - \alpha r - (1 - \mu)(1 - \alpha) \right].\end{aligned}\tag{23}$$

Expanding the bracket:

$$V_e(\mu) - \alpha r - (1 - \mu)(1 - \alpha) = \mu(r - \alpha) - \alpha(r - 1),$$

so $\delta_{R2}(\mu) = (N-1)\beta[\mu(r - \alpha) - \alpha(r - 1)]/N^2$, which matches the formula in Appendix B.5. At $\mu = \mu_s^{R2} = \alpha(r - 1)/(r - \alpha)$, direct substitution gives $\delta_{R2}(\mu_s^{R2}) = 0$, confirming continuity at the R2 cutoff.

Step 4: Additivity. With both corrections in hand, I can now establish that they combine additively. This is the key structural result: it means the full payoff on any branch of the belief space can be written as the benchmark plus whichever corrections are active, without interaction terms.

The two corrections are independent because they affect disjoint components of H 's payoff:

- δ_{R2} operates through Π_H^{prop} (equation (15)), which depends on $V_W^{R2}(\mu)$ but is invariant to the R1 regime.
- δ_{R1} operates through Π_H^W (equation (18)), which depends on the R1 offer type but is invariant to $V_W^{R2}(\mu)$ —because the R1 offers to H are pinned at $\beta V_H^{R2}(\theta, \mu' = 1)$, which is always

evaluated at $\mu' = 1 > \mu_s^{R2}$.

Hence the full unanimity payoff is:

$$E[V_H^{R1}(\mu, U)] = E_{\text{bench}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

Step 5: Subtracting majority and collecting terms. The final step subtracts the majority payoff from the unanimity benchmark to obtain D_{base} . This completes the decomposition: the payoff difference $D(\mu)$ on any branch equals D_{base} plus the active corrections, and all three components are affine in μ .

Under majority (Appendix A.1), $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1 + (N - 1)\alpha) - C_{\text{buy}}]/N^2$ with $C_{\text{buy}} = \beta(q - 1)(1 - \alpha)$ and $q = \lfloor N/2 \rfloor + 1$. Define $C_{\text{out}} = N(N - 1)\alpha$. Then:

$$\begin{aligned} D_{\text{base}}(\mu) &\equiv E_{\text{bench}}(\mu) - \lambda_M V_e(\mu) \\ &= \frac{N V_e(\mu) + (N - 1)\beta N \alpha r}{N^2} - \frac{N(1 + (N - 1)\alpha) - C_{\text{buy}}}{N^2} V_e(\mu) \\ &= \frac{V_e(\mu)[N - N - C_{\text{out}} + C_{\text{buy}}] + C_{\text{out}}\beta r}{N^2} \\ &= \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \end{aligned} \tag{24}$$

where the third line uses $N(1 + (N - 1)\alpha) = N + C_{\text{out}}$ and $(N - 1)\beta N \alpha r = C_{\text{out}}\beta r$.

The payoff difference is therefore:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu),$$

with D_{base} , δ_{R2} , δ_{R1} as in equations (24), (23), (22). $\square$

B.5 Proof of Theorem 1 (Conditional payoff dominance)

The payoff difference $D(\mu)$ decomposes into a baseline affine term plus two correction terms, each itself affine. Positivity therefore reduces to checking endpoints. The decomposition is derived in Appendix B.5a.

Let $C_{\text{buy}} \equiv \beta(q-1)(1-\alpha)$ (the vote-buying cost under majority) and $C_{\text{out}} \equiv N(N-1)\alpha$ (the aggregate outside-option cost). Under majority, $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1 + (N-1)\alpha) - C_{\text{buy}}]/N^2$.

Step 1: Decomposition. The decomposition has three components, each affine in μ . Because an affine function that is positive at both endpoints of an interval is positive throughout, it suffices to check a finite number of boundary values.

Define:

$$D_{\text{base}}(\mu) \equiv \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \quad (25)$$

$$\delta_{R2}(\mu) \equiv \frac{(N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]}{N^2}, \quad (26)$$

$$\delta_{R1}(\mu) \equiv \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \quad (27)$$

The difference $D(\mu) \equiv E[V_H^{R1}(\mu, U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

This decomposition holds regardless of the relative ordering of μ_s^{R1} and μ_s^{R2} . The two corrections are additive because they affect independent components of H 's payoff: δ_{R2} captures the change in $V_W^{R2}(\mu)$ (which enters through H 's proposal offer), while δ_{R1} captures the change in the R1 offer H receives when W proposes (which depends on the R1 regime but not on V_W^{R2}).

D_{base} is the difference under aggressive R1 and conservative R2. The R2 correction vanishes at the screening cutoff: $\delta_{R2}(\mu_s^{R2}) = 0$, ensuring continuity at μ_s^{R2} . The R1 correction satisfies $\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$, with equality only at $\mu = 1$.

The proof strategy is as follows. D_{base} is positive everywhere on $[0, 1]$ (Step 2). The R2 correction δ_{R2} is active only below μ_s^{R2} ; it is negative at $\mu = 0$ but vanishes at μ_s^{R2} , so the tighter endpoint condition is $D_I(0) = D_{\text{base}}(0) + \delta_{R2}(0) > 0$. The R1 correction δ_{R1} is active only above μ_s^{R1} and is non-negative throughout, so it can only help. Step 3 assembles these facts branch by branch.

Step 2: $D_{\text{base}} > 0$ on $[0, 1]$. Since D_{base} is affine in μ —it depends on μ only through $V_e(\mu) =$

$1 + \mu(r - 1)$, which is itself affine—checking positivity at the two endpoints $\mu = 0$ and $\mu = 1$ is both necessary and sufficient.

At $\mu = 1$:

$$D_{\text{base}}(1) = \frac{r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]}{N^2}.$$

Setting $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$ and solving for α yields $\alpha = \alpha^*(N, \beta)$. Since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$.

At $\mu = 0$: Define $D_I(\mu) \equiv D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, the payoff difference on the low-belief branch. We check $D_I(0)$ rather than $D_{\text{base}}(0)$ directly, because $\delta_{R2}(0) < 0$ —so $D_I(0)$ is the tighter endpoint condition. Write $D_I(0) = [\beta(q - 1) - \alpha d_0]/N^2$ where $d_0 = N(N - 1) - \beta((N - 1)^2 r + N - q)$. If $d_0 \leq 0$, then $D_I(0) \geq \beta(q - 1)/N^2 > 0$. If $d_0 > 0$, the threshold $\bar{\alpha}_0 = \beta(q - 1)/d_0$ satisfies $\bar{\alpha}_0 > \alpha^*$ because $d_0 < d_* \equiv N(N - 1) - \beta(N^2 - N - q + 1)$ (since $d_* - d_0 = \beta(N - 1)^2(r - 1) > 0$). Hence $\alpha < \alpha^* < \bar{\alpha}_0$ implies $D_I(0) > 0$, and therefore $D_{\text{base}}(0) > D_I(0) > 0$.

Conclusion: D_{base} is affine with positive values at both endpoints, hence $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 3: Assembly. The strategy is to partition the belief space $(0, 1]$ by the two screening cutoffs and verify $D > 0$ on each branch, using the endpoint results from Steps 1–2. On every branch, D is a sum of affine functions and is therefore itself affine, so positivity at branch endpoints implies positivity throughout.

The standard case is $\mu_s^{R2} \leq \mu_s^{R1}$, which partitions $(0, 1]$ into three branches.

Low-belief branch ($\mu < \mu_s^{R2}$): $D = D_I(\mu)$, which is affine. At the left endpoint, $D_I(0) > 0$ (Step 2). At the right endpoint, $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2). Affine with positive endpoints implies $D > 0$ throughout.

Middle branch ($\mu_s^{R2} \leq \mu \leq \mu_s^{R1}$): $D = D_{\text{base}} > 0$ directly from Step 2.

High-belief branch ($\mu > \mu_s^{R1}$): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Alternative case ($\mu_s^{R1} < \mu_s^{R2}$). The additive decomposition remains valid because the two corrections affect independent components of H 's payoff (Step 1). The partition of $(0, 1]$ now has three branches:

Low-belief branch ($\mu < \mu_s^{R1}$): aggressive R1, aggressive R2. $D = D_I(\mu) = D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, which is affine. At $\mu = 0$, $D_I(0) > 0$ (Step 2). At $\mu = \mu_s^{R1}$, D_I is continuous from the left and $D_I(\mu_s^{R1}) > 0$ because D_I is affine with $D_I(0) > 0$ and $D_I(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2), and $\mu_s^{R1} < \mu_s^{R2}$.

Middle branch ($\mu_s^{R1} \leq \mu < \mu_s^{R2}$): conservative R1, aggressive R2. Both corrections are active: $D = D_{\text{base}} + \delta_{R2} + \delta_{R1}$. This is affine. At μ_s^{R1} , D jumps up by $\delta_{R1}(\mu_s^{R1}) > 0$ relative to the low branch, so $D(\mu_s^{R1}) = D_I(\mu_s^{R1}) + \delta_{R1}(\mu_s^{R1}) > D_I(\mu_s^{R1}) > 0$. At μ_s^{R2} , $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) + \delta_{R1}(\mu_s^{R2}) \geq D_{\text{base}}(\mu_s^{R2}) > 0$. Affine with positive values at both endpoints implies $D > 0$ throughout.

High-belief branch ($\mu \geq \mu_s^{R2}$): conservative R1, conservative R2. $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Combining all branches in either case: $D(\mu) > 0$ for every $\mu \in (0, 1]$. This establishes sufficiency.

Step 4: Necessity. To establish that $\alpha < \alpha^*$ is not merely sufficient but also necessary, I show that the inequality $D(\mu) > 0$ fails at $\mu = 1$ whenever $\alpha \geq \alpha^*$.

At $\mu = 1$, both corrections vanish ($\delta_{R1}(1) = 0$ and $\mu = 1 > \mu_s^{R2}$), so $D(1) = D_{\text{base}}(1) = r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]/N^2$. Since α^* is the root of $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$, $\alpha \geq \alpha^*$ implies $D(1) \leq 0$, and the strict inequality fails. $\square$

B.6 Proof of Corollary (Unanimity dominance on the formation set)

Fix $p \in \mathcal{F}_U$, so $V_W^{R1}(p, U) \geq c$. We first show that $p \in \mathcal{F}_M$. Under majority, the budget identity holds exactly: $E[V_H(\mu, M)] + (N-1)V_W(\mu, M) = V_e(\mu)$. Under unanimity, surplus destruction from discounting on the aggressive branch yields $E[V_H(\mu, U)] + (N-1)V_W(\mu, U) \leq V_e(\mu)$. Since Theorem 1 gives $E[V_H(\mu, U)] > E[V_H(\mu, M)]$, it follows that $V_W(\mu, U) < V_W(\mu, M)$ for every $\mu \in (0, 1]$. Hence $V_W^{R1}(p, M) > V_W^{R1}(p, U) \geq c$, so entry occurs under majority at p . In particular, $\mathcal{F}_U \subseteq \mathcal{F}_M$.

Since the institution forms under both rules at p , the hegemon's payoff under each rule is the expected bargaining payoff: $V_H(R, p) = E_\theta[V_H^{R1}(\theta, p, R)]$. By Theorem 1, $V_H(U, p) > V_H(M, p)$. $\square$

B.7 Lemma: Global maximum of V_W^{R1} under unanimity

Lemma 1 (Global maximum of weak-state payoff). $V_W^{R1}(\mu, U) \leq V_W^{R1}(1, U) = r(1 - \beta\alpha)/N$ for every $\mu \in (0, 1]$, with equality only at $\mu = 1$. Hence if $\mathcal{F}_U \neq \emptyset$, then $1 \in \mathcal{F}_U$.

Proof. At $\mu = 1$, the equilibrium R1 offer is conservative: $F_1^{con}(1) = r - \beta(r + x)/N - \omega(1) > 0$ while $F_1^{agg}(1) = \beta r(1 - \alpha)/N > 0$, and the conservative surplus exceeds the aggressive surplus because $\beta < 1$ implies the full pie exceeds the discounted R2 value. Hence $\bar{V}_W \equiv V_W^{R1}(1, U) = r(1 - \beta\alpha)/N$.

We bound every feasible R1 payoff candidate by $\bar{V}_W$, using the full ex ante weak-state payoff (not only the proposer surplus). Let $W_2^L(\mu) = (1 - \mu)(1 - \alpha)/N$ and $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ denote the R2 continuation values on the low and high branches, $W_2^1 = r(1 - \alpha)/N$ the R2 value at $\mu = 1$, and $x = (N - 1)\alpha r$. Under a conservative R1 offer (all types accept), a non-proposing weak state receives $\beta W_2^j(\mu)$ whenever another player proposes, giving $V_W^{Cj}(\mu) = F_1^{con,j}(\mu)/N + (N - 1)\beta W_2^j(\mu)/N$. Under an aggressive R1 offer ($\theta = 1$ rejects, game proceeds to R2 with posterior one), $V_W^{Aj}(\mu) = F_1^{agg,j}(\mu)/N + \beta W_2^j(\mu)/N + (N - 2)[(1 - \mu)\beta W_2^j(\mu) + \mu\beta W_2^1]/N$. The equilibrium payoff $V_W^{R1}(\mu, U)$ is one of the four candidates V_W^{CH} , V_W^{AH} , V_W^{CL} , V_W^{AL} , depending on the R1 offer type and the R2 branch.

Conservative R1, high R2 ($\mu \geq \mu_s^{R2}$): $V_W^{CH}(\mu) = [(N + \beta)V_e(\mu) - \beta r(1 + N\alpha)]/N^2$, so $\bar{V}_W - V_W^{CH}(\mu) = (N + \beta)(r - 1)(1 - \mu)/N^2 \geq 0$, with equality only at $\mu = 1$.

Aggressive R1, high R2 ($\mu \geq \mu_s^{R2}$): The proposer surplus $F_1^{agg,H}$ contains a term $(1 - \mu)\beta W_2^H(\mu)$ that is quadratic in μ , but this term cancels exactly with the corresponding non-proposer term $(1 - \mu)\beta W_2^H(\mu)$ in V_W^{AH} , leaving a result that is affine. After substituting $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ and simplifying: $\bar{V}_W - V_W^{AH}(\mu) = [(1 - \mu)(r - 1) + \mu r(1 - \beta)]/N > 0$ for all $\mu \in [0, 1]$.

Conservative R1, low R2 ($\mu < \mu_s^{R2}$): Using $W_2^L(\mu) = (1 - \mu)(1 - \alpha)/N$, $\bar{V}_W - V_W^{CL}(\mu)$ is affine in μ . At $\mu = 0$: $(r - 1)(N - \alpha\beta + \beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r - 1)(1 - \alpha)(N + \beta)/[N^2(r - \alpha)] > 0$. Affine with positive endpoints implies $\bar{V}_W - V_W^{CL} > 0$ on $[0, \mu_s^{R2}]$.

Aggressive R1, low R2 ($\mu < \mu_s^{R2}$): The same quadratic cancellation as in the high-R2 case applies: the $(1 - \mu)\beta W_2^L(\mu)$ terms in the proposer and non-proposer components cancel, and

$\bar{V}_W - V_W^{AL}(\mu)$ is affine. At $\mu = 0$: $(r - 1)(N - \alpha\beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r - 1)(1 - \alpha\beta)/[N(r - \alpha)] > 0$. Hence $\bar{V}_W - V_W^{AL} > 0$ on $[0, \mu_s^{R2}]$.

Since every payoff candidate is strictly below $\bar{V}_W$ for $\mu < 1$, and the equilibrium payoff selects one of these candidates, $V_W^{R1}(\mu, U) < \bar{V}_W$ for all $\mu < 1$. The corollary follows: if $V_W^{R1}(\mu', U) \geq c$ for some μ' , then $V_W^{R1}(1, U) \geq V_W^{R1}(\mu', U) \geq c$, so $1 \in \mathcal{F}_U$. $\square$

B.8 Proof of Proposition 4 (Institutional classification)

We verify each case exhaustively. Since $\mathcal{F}_U \subseteq \mathcal{F}_M$ (Corollary, Appendix B.6), the three cases $p \in \mathcal{F}_U$, $p \in \mathcal{F}_M \setminus \mathcal{F}_U$, and $p \notin \mathcal{F}_M$ partition $(0, 1]$.

Case (i): $p \in \mathcal{F}_U$. The institution forms under both rules. By Corollary~1, $V_H(U, p) > V_H(M, p)$, so $U \succ M$.

Case (ii): $p \in \mathcal{F}_M \setminus \mathcal{F}_U$. The institution forms under majority but not under unanimity. Under unanimity, $V_H(U, p) = \alpha V_e(p)$ (the bilateral outside option). Under majority, the hegemon's payoff from the institution exceeds the outside option: $V_H(M, p) = E_\theta[V_H^{R1}(\theta, p, M)] > \alpha V_e(p)$, since entry is individually rational for weak states (which requires positive surplus above outside options). Hence $M \succ U$.

Case (iii): $p \notin \mathcal{F}_M$. The institution does not form under either rule (since $\mathcal{F}_U \subseteq \mathcal{F}_M$, $p \notin \mathcal{F}_M$ implies $p \notin \mathcal{F}_U$). Both rules yield the outside-option payoff $\alpha V_e(p)$, so $U \sim M$. $\square$

Appendix C: Extension to $K > 2$ Types

This appendix examines how the screening mechanism and the majority-linearity result extend to $K > 2$ types. The majority-linearity result generalizes fully (Proposition 5). The screening structure under unanimity becomes richer— $K - 1$ boundaries rather than one—but whether unanimity dominates majority for empirically relevant parameterizations with $K \geq 3$ is not established here.

C.1 Setup

The model is identical to Section 4 except that $\theta \in \{0, 1, 2\}$ with $V(0) = 1$, $V(1) = r_1$, $V(2) = r_2$, where $1 < r_1 < r_2$. The prior is $p = (p_0, p_1, p_2) \in \Delta^2$, the 2-simplex. The expected value of cooperation is $V_e(\mu) = \sum_k \mu_k V(\theta_k)$. All other primitives— N players, discount factor β , random proposer, outside option $d_H = \alpha V(\theta)$ —are unchanged. The constraint on the outside option becomes $\alpha \in (0, 1/r_2)$, ensuring that $d_H < V(\theta)$ for all types.

C.2 R2 Screening under Unanimity

With K types, the R2 screening problem has a natural generalization. When W makes an offer in R2, it targets a particular type by setting the reservation value equal to $\alpha V(\theta_k)$ for some k . All types θ_j with $j \leq k$ accept (since $\alpha V(\theta_j) \leq \alpha V(\theta_k)$), and all types θ_j with $j > k$ reject. The optimal target balances the probability of acceptance against the cost of overpaying low types.

Claim (R2 screening boundaries for general K). *For K ordered types with values $v_1 < v_2 < \dots < v_K$, the R2 screening boundary between targeting type k and type $k + 1$ is defined by the posterior ratio*

$$\frac{\mu_{k+1}}{\sum_{j \leq k} \mu_j} = \frac{\alpha(v_{k+1} - v_k)}{v_{k+1}(1 - \alpha)}.$$

These $K - 1$ conditions define hyperplanes that partition the simplex Δ^{K-1} into K screening regions.

The logic is as follows. The weak proposer's payoff from an offer targeting type k is $\sum_{j \leq k} \mu_j [V(\theta_j) - \alpha V(\theta_k)]$. Equating the payoffs from targeting type k and type $k + 1$ yields the indifference condition. The ratio simplifies because the incremental gain from including type $k + 1$ (at cost $\alpha V(\theta_{k+1})$) must equal the incremental cost of overpaying all types $j \leq k$ (the additional payment $\alpha[V(\theta_{k+1}) - V(\theta_k)]$ per accepting type). For $K = 2$, this reduces to $\mu_s^{R2} = \alpha(r - 1)/(r - \alpha)$, recovering the main model.

For $K = 3$ with values $(1, r_1, r_2)$, the R2 screening produces three regions separated by two boundaries. The continuation values for H in R2 follow the same logic as the binary case: under a high offer (targeting $\theta = 2$), types $\theta = 0$ and $\theta = 1$ are overpaid; under a medium offer (targeting $\theta = 1$), only $\theta = 0$ is overpaid; under a low offer (targeting $\theta = 0$), all types receive their outside

option. The key pattern persists: low types receive strictly more than their outside option whenever W targets a higher type, and the highest type $\theta = 2$ always receives $r_2(1 + (N-1)\alpha)/N$ regardless of the screening region.

C.3 Majority: No Screening

Proposition 5 (Linearity of H 's Payoff under Majority for General K). *Under majority rule with K types, $E_\theta[V_H^{R1}(\theta, \mu, M)] = \lambda_M V_e(\mu)$, where λ_M is defined as in Appendix B.5. No screening cutoff exists for any K .*

Proof. The proof is identical to the $K = 2$ case. Under majority, W never needs H 's vote; H receives $\alpha V(\theta)$ in every coalition that forms and $\alpha V(\theta)$ as its outside option. Since H 's payoff is $\alpha V(\theta)$ type by type, the expectation is $\alpha V_e(\mu)$ scaled by the bargaining multiplier λ_M/α . The belief μ enters only through $V_e(\mu)$, so the payoff is affine in μ (and hence in V_e) regardless of K . $\square$

Together, the screening claim above and Proposition 5 establish the structural asymmetry for general K : unanimity generates $K - 1$ screening boundaries, each generating a conditional payoff advantage for the hegemon, while majority remains linear regardless of K .

Limitations and continuous-type extension. For $K = 2$, Theorem 1 provides a sharp threshold α^* below which unanimity conditionally dominates at every belief. For discrete $K \geq 3$, the analogous threshold α_K^* is harder to characterize: with more types, the vote-buying cost of unanimous agreement at the highest type grows, potentially tightening the parametric region where dominance holds. What generalizes without qualification is the structural contrast: unanimity creates screening, majority does not (Proposition 5).

C.4 Continuous types

The screening mechanism extends to continuous type distributions. Suppose $\theta \sim F$ on $[1, r]$ with full support. Under unanimity, when W proposes, W faces a screening problem: choosing a cutoff θ^* such that types $\theta \leq \theta^*$ accept the offer $\alpha\theta^*$ and types $\theta > \theta^*$ reject. The screening rent—the

excess payoff to H relative to the perfect-discrimination benchmark $\alpha\theta$ —is

$$\text{Rent} = \int_1^{\theta^*} \alpha(\theta^* - \theta) dF(\theta) > 0$$

for any equilibrium cutoff $\theta^* > 1$. Under majority, W bypasses H , and the rent is zero. This holds for any continuous F with full support. The qualitative mechanism is distribution-free.

For the uniform distribution $F = \text{Uniform}[1, r]$, the full conditional payoff comparison admits a closed-form solution. The R2 screening cutoff is $\theta_2^* = r$ (all types are pooled), which eliminates the piecewise structure and makes all continuation values linear in θ . The payoff difference $D = E_\theta[V_H^{R1}(U)] - E_\theta[V_H^{R1}(M)]$ is a polynomial in (α, r, N, β) , and the continuous-type dominance threshold α_{cont}^* satisfies $\alpha_{\text{cont}}^* \geq \alpha^*$ for all parameter values, with strict inequality whenever $r > 1$. The binary model is therefore the most conservative case for the mechanism: continuous types generate richer overpayment because more types fall below the screening cutoff, amplifying the aggregate screening rent. Numerical verification across 1,008 parameter combinations confirms zero violations.

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